

DEEKSHA MALLAMPET

DATA SCIENTIST – HEALTHCARE | ML ENGINEERING | MLOPS

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PROFESSIONAL SUMMARY

DATA SCIENTIST WITH 3+ YEARS OF EXPERIENCE IN HEALTHCARE ANALYTICS, CLOUD-BASED DATA ENGINEERING, AND REPORTING AUTOMATION. ADEPT AT BUILDING SCALABLE ETL PIPELINES, IMPLEMENTING HIPAA-COMPLIANT WORKFLOWS, AND ENABLING BUSINESS INTELLIGENCE VIA PYTHON, SQL, TABLEAU, PYSPARK, AND AWS. PROVEN ABILITY TO TRANSFORM MESSY REAL-WORLD HEALTHCARE DATA INTO ACTIONABLE INSIGHTS THAT ENHANCE OPERATIONS, REDUCE COSTS, AND SUPPORT CLINICAL DECISION-MAKING. PASSIONATE ABOUT SOLVING PROBLEMS AT THE INTERSECTION OF DATA, HEALTHCARE, AND PATIENT EXPERIENCE.

CORE SKILLS

PROGRAMMING & ANALYTICS: PYTHON (PANDAS, NUMPY, SCIKIT-LEARN), SQL, PYSPARK, SHELL, JUPYTER NOTEBOOKS
MACHINE LEARNING & AI: REGRESSION, CLASSIFICATION, CLUSTERING, A/B TESTING, FEATURE ENGINEERING, XGBOOST, TENSORFLOW
DEEP LEARNING & NLP: CNNs, RNNs, LSTMS, BERT, TRANSFORMERS, VISION-LANGUAGE MODELS, TRANSFER LEARNING
DATA ENGINEERING & ETL: SNOWFLAKE, AIRFLOW, APACHE KAFKA, SSIS, AWS GLUE, DATA CLEANING & TRANSFORMATION
MLOPS & DEPLOYMENT: DOCKER, JENKINS, GITHUB ACTIONS, CI/CD PIPELINES, AWS SAGEMAKER, MLFLOW, DVC
BUSINESS INTELLIGENCE & VISUALIZATION: TABLEAU, POWER BI, PLOTLY, SEABORN, DASH, STREAMLIT
CLOUD PLATFORMS: AWS (S3, EC2, LAMBDA, GLUE), AZURE (FUNCTIONS, BLOB STORAGE), GCP (BIGQUERY)
MONITORING & MODEL GOVERNANCE: MLFLOW, PROMETHEUS, GRAFANA, DATA LINEAGE, MODEL VERSIONING
TOOLS & IDEs: JUPYTER, VS CODE, GITHUB, EXCEL, LINUX, GOOGLE COLAB
METHODOLOGIES & COLLABORATION: AGILE, CRISP-DM, SDLC, HIPAA COMPLIANCE, JIRA, MS TEAMS, SLACK

PROFESSIONAL EXPERIENCE

CIGNA – MASSACHUSETTS, USA

DATA SCIENTIST | JAN 2025 – PRESENT

- BUILT SCALABLE ETL PIPELINES WITH PYTHON AND SQL TO INGEST >20M PATIENT RECORDS FROM AWS S3 INTO MYSQL, IMPROVING AVAILABILITY BY 40%.
- DESIGNED TABLEAU DASHBOARDS TO TRACK INVENTORY TURNOVER, SALES KPIS, AND PATIENT UTILIZATION TRENDS - REDUCING DECISION LATENCY BY 30%.
- AUTOMATED WEEKLY REPORTING WORKFLOWS WITH SSIS AND SQL SCRIPTS, REDUCING MANUAL LABOR BY 60% AND IMPROVING OPERATIONAL EFFICIENCY.
- CONDUCTED FEATURE ENGINEERING AND DATA NORMALIZATION USING PANDAS AND SCIKIT-LEARN, REDUCING DOWNSTREAM MODEL ERRORS BY 20%.
- ENSURED HIPAA COMPLIANCE THROUGH RIGOROUS DATA VALIDATION AND QA CHECKS ACROSS ALL PROCESSED DATASETS.
- EXECUTED BATCH PROCESSING ON AWS EC2 TO SCALE PIPELINE WORKLOADS AND REDUCE INFRASTRUCTURE OVERHEAD.
- WORKED IN AGILE/SCRUM CROSS-FUNCTIONAL PODS TO TRANSLATE ANALYTICS INTO ACTIONABLE INSIGHTS.

NEON IT SYSTEMS – HYDERABAD, INDIA

DATA SCIENTIST | JAN 2020 – JULY 2022

- DEVELOPED BATCH AND STREAMING ETL PIPELINES USING PYSPARK AND SNOWFLAKE TO PROCESS HEALTHCARE CLAIMS AND INTERACTION LOGS, REDUCING LATENCY BY 35%.
- WROTE OPTIMIZED SQL QUERIES AND PROCEDURES TO SUPPORT BI DASHBOARDS, CUTTING DATA RETRIEVAL TIME BY 25%.
- INTEGRATED AZURE FUNCTIONS TO TRIGGER ENRICHMENT TASKS (E.G., JSON PARSING), ELIMINATING 30% VM COST OVERHEAD.
- BUILT INTERACTIVE DASHBOARDS USING PLOTLY AND TABLEAU FOR MARKETING, PATIENT CHURN, AND REVENUE TRENDS ANALYSIS.
- AUTOMATED CI/CD FOR ETL DEPLOYMENT USING JENKINS AND GITHUB, MAINTAINING DATA RELIABILITY AND VERSION CONTROL.
- ENSURED DATA QUALITY THROUGH LINEAGE TRACKING AND AIRFLOW METADATA PIPELINES IN THE SNOWFLAKE DATA WAREHOUSE.
- DELIVERED AD HOC BUSINESS REPORTS TO EXECUTIVES USING SQL AND EXCEL/CSV EXPORTS UNDER STRICT SLAS.

ACADEMIC PROJECTS

CONTEXT-AWARE IMAGE CAPTIONING USING LLM + VLM

UNIVERSITY OF MASSACHUSETTS DARTMOUTH

- DEVELOPED A HYBRID LLM-VLM FRAMEWORK USING PSEUDO-LABELING AND PROMPT OPTIMIZATION TO IMPROVE IMAGE CAPTION ACCURACY ON THE WHOOPS DATASET (500 SAMPLES).
- INTEGRATED BERT EMBEDDINGS AND ITERATIVE FEEDBACK FOR SEMANTIC REFINEMENT.
- EVALUATED CAPTIONS USING BLEU SCORES AND ENSURED REPRODUCIBILITY IN JUPYTER LAB WITH PYTHON-BASED PIPELINES.

CAR PRICE PREDICTION

UNIVERSITY OF MASSACHUSETTS DARTMOUTH

- BUILT A REGRESSION MODEL IN SCIKIT-LEARN TO PREDICT CAR PRICES USING TECHNICAL SPECS AND BODY FEATURES.
- PERFORMED EDA TO UNCOVER PRICING DRIVERS AND TRENDS.
- DESIGNED TABLEAU DASHBOARDS TO VISUALIZE PRICING INSIGHTS AND RECOMMENDATIONS.

EDUCATION

MASTER OF SCIENCE – DATA SCIENCE

UNIVERSITY OF MASSACHUSETTS DARTMOUTH | MAY 2025

BACHELOR OF ENGINEERING – ELECTRONICS & COMMUNICATION ENGINEERING

MALLAREDDY ENGINEERING COLLEGE, HYDERABAD, INDIA | JUNE 2023

CERTIFICATIONS

- TABLEAU FOR DATA SCIENCE – COURSERA
- ADVANCED SQL FOR DATA ANALYTICS – UDEMY
- POWER BI DASHBOARDING – LINKEDIN LEARNING